Ratul Hasan

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EDUCATION

Bangladesh Army University of Engineering & Technology (BAUET) — *Natore, Bangladesh*
B.Sc in Computer Science and Engineering | CGPA: 3.82/4.0 — *2022 – 2026*

Thesis Title: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimer's Disease

Govt. Majid Memorial City College — *Khulna, Bangladesh*
Higher Secondary School Certificate | GPA: 5.0/5.0 — *2019 – 2021*

EXPERIENCE

ELITE Research Lab LLC | *Remote* — *Queens, NY, USA*

- ***Research Assistant*** — *October 2026 - Present*
  - Working on a team journal project mapping attacker-controllable IoT metadata as an indirect prompt-injection surface for LLM home agents
  - Measuring which Home Assistant device and integration text fields reach its built-in LLM conversation agents, and how often injected instructions trigger unauthorised physical actions or leak private state in a simulated home
  - Evaluating how provenance tagging, actuating-tool gating and action confirmation reduce attack success, and their cost in benign-task utility
- ***Research Intern*** — *June 2026 - Sept 2026*
  - First-authored a systematic review of 82 studies on LLMs in mental health, appraising methodological rigour and risk reporting (hallucination, bias, safety, human evaluation) with a 24-field extraction instrument
  - Studied catastrophic forgetting in federated continual learning for mental health text classification, separating label-space shift from domain shift on Reddit stress and depression data
  - Benchmarked seven training regimes (FedAvg, FedProx, EWC, replay, centralized and joint baselines) across 10 clients and three seeds with a MiniLM encoder

BAUET Computer Society, Department of CSE, BAUET | ***Programming Instructor*** — *Natore, Bangladesh*
July 2024 - December 2025

- Conducted programming classes for undergraduate students covering C, C++ and basic data structures.
- Led coding exercises and programming contest preparation sessions to strengthen students' problem-solving and analytical skills.
- Explained programming concepts to students of varying skill levels and supported the development of their programming fundamentals.

Lab AR | ***Software Engineering Intern*** — *Dhaka, Bangladesh*
July 2025 - August 2025

- Led a team project to develop a web-based Property Management System.
- Contributed to backend development using Laravel (PHP) and collaborated on feature implementation.

PUBLICATIONS

Conference Publications

Quantum-Resistant FOTA: End-to-End Decentralized Firmware Updates for IoT Using Blockchain and CRYSTALS-Dilithium — *2025*

Ratul Hasan, Md. Momenul Haque, Redoanul Haque, Yousuf Oley and Ruhani Akter. *Proceedings of the 12th International Conference on Next Generation Computing, Communication, Systems and Security (NSysS '25).* ACM, New York, NY, USA, 110–114. doi.org/10.1145/3777555.3777569

A Gas-Optimized Blockchain Framework for Scalable Document Verification Using Dynamic Access Control and IPFS — *2026*

Ratul Hasan, Samiha Farjana, Yousuf Oley, Md. Ohiduzaman Pranto and Md Arik Rayhan. Published in the *International Conference on Power, Electronics, Communications, Computing and Intelligent Infrastructure (PECCII 2026)*, IEEE, 2026. https://doi.org/10.1109/PECCII70991.2026.11662071

SplitSafe Ultra: A Moving-Target Defense Framework for Entropy-Aware Encrypted File Persistence on IPFS — *Expected 2026*

Md. Ahnaf Muhaimin, **Ratul Hasan**, Yousuf Oley, Samiha Farjana, Bristi Rani Roy, Subrata Kumer Paul and Md. Ekramul Hamid. Accepted to the *5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026).* To be published in IEEE Proceedings.

ActiVisionNet-Fusion: A Dual-Stream Explainable Deep Learning Framework for Driver Drowsiness Detection *Expected 2026*

Nasim Ahmed, **Ratul Hasan**, Subrata Kumer Paul, Dewan Nafiul Islam Noor, Md. Momenul Haque and Md. Ekramul Hamid. Accepted to the *5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)*. To be published in IEEE Proceedings.

Journal Publications

BI-RNN: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimer's Disease *Under Review*

Ratul Hasan, Md. Momenul Haque, Ananya Sarker. Under review in *Nature Scientific Reports*, Submission Date: May 2026.

Entropy-Assured Post-Quantum Key Generation on IoT Edge Devices: Source Independence, Measured Min-Entropy and Signature Scheme Selection *Under Review*

Ratul Hasan, Md Arik Rayhan and Md Ohiduzaman Pranto. Under review in *PLOS ONE*.

COMPETITIVE PROGRAMMING & OTHERS

Online Judges

- Codeforces: Max Rating 1650 (Expert) **Ratul_Hasan**
- Codechef: Max Rating 1676 (3 Star) **ratulhasan2108**
- Leetcode: Solved more than 50+ problems **ratulhasan001**
- Solved more than 1000+ problems across various online judges

Programming Contests

- 1X ICPC Regionalist (Team: BAUET_Rong_Answer)
- 1X National Collegiate Programming Contest by JU (Rank: 149, Team: BAUET_PARADIGM)
- 2X Inter University Programming Contest (Best Rank: 45 at IUT, Team: BAUET_Twisted_Minds)
- Ranked 10th in Independence Day Programming Contest by Military Institute of Science & Technology
- 1X Champion, 1X 2nd runners-up in Intra University Programming Contest (Team)
- 1X Runners-up in Intra University Programming Contest (Solo)

Others

- 1X 1st Runners Up in Intra University Three Minutes Thesis Presentation

PROJECTS

LedgerSeal: A Blockchain Based Document Verification Framework using Dynamic Access Control and IPFS *Github*

- Developed a decentralized application (DApp) for constant-time document verification with dynamic access control on IPFS using a public blockchain.
- Tools Used: React JS, Anvil EVM, Foundry, Metamask, Pinata IPFS

BAUET Project & Thesis Archive *Github*

- Developed a repository website using Django(Both MVT & REST) where theses and projects of BAUET are represented
- Tools Used: HTML, Tailwind CSS, JavaScript, Django, PostgreSQL

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL || **Frameworks:** Django, REST, Foundry, Laravel
Project Management: Git, Github || **Database:** MySQL, PostgreSQL
Technologies: Machine Learning, Computer Vision, Backend Development, Blockchain

VOLUNTEERING & LEADERSHIP EXPERIENCE

- President, BAUET Computer Society (Advanced through five executive leadership positions)
- Problem Setter, Intra-University Programming Contest, BAUET
- Officer of Documentation, Hult Prize at BAUET